

Name: MRS. INDU SINGH
Ref By: DR. MOHAN KESHAVAMURTHY
ID: 13822742

Age: 56 Y
Sex: F
Date: 17.05.2025

Head & Neck:

The neuroparenchyma is unremarkable; no focal mass lesion/mass effect. The mastoid air cells and other paranasal sinuses are unremarkable. The parotid and submandibular glands are unremarkable. *Prominent FDG uptake is noted in bilateral tonsils- likely reactive.* The pharyngeal mucosal space, parapharyngeal, retropharyngeal, posterior cervical, carotid and perivertebral spaces are unremarkable. The larynx and cervical trachea are unremarkable. The thyroid gland is unremarkable. *Supraclavicular lymph node as described above.*

Chest:

Bibasilar dependent atelectasis is noted in the lungs; *lung nodules and mediastinal lymph nodes as described above.* No evidence of pleural effusion or pneumothorax. Lower trachea and main bronchi are unremarkable. Esophagus is unremarkable. The cardia is normal; no pericardial effusion. Bilateral breasts and axillae are unremarkable.

Abdomen and pelvis:

Liver: Findings as described above. No IHBR dilatation. Portal vein, hepatic veins and intrahepatic IVC are normal.

Gall bladder: Unremarkable. The CBD is not dilated.

Spleen: Unremarkable. Pancreas: Unremarkable. Adrenal glands: Findings as described above.

Kidneys: Findings as described above.

G.I tract: The stomach is partially distended; duodenum, jejunum and ileum are unremarkable; colon and rectum are loaded with fecal matter and air.

Peritoneum and retroperitoneum: No free fluid, fluid collection or free air in the peritoneum. *Retroperitoneal lymph nodes as described above.*

Pelvis: Uterus and ovaries are unremarkable. Urinary bladder is partially distended and otherwise unremarkable. No significant pelvic lymphadenopathy. No significant inguinal lymphadenopathy.

Musculoskeletal system:

Degenerative changes are noted in the spine; no abnormal focal osteolytic or osteosclerotic or hypermetabolic lesions.

Vascular Structures:

Atheromatous changes are seen in aorta; major vessels in the neck, chest and abdomen are otherwise unremarkable.

IMPRESSION:

- The PETCT study shows metabolically active lesion in the upper pole cortex of left kidney, asymmetrical circumferential mural thickening in the pelvicalyceal system extending to proximal ureter with lymph nodes (retroperitoneal, mediastinal and left supraclavicular), lung nodules and liver lesion as described above - Likely representing TCC left kidney - proximal ureter with metastasis. Recommend further biopsy and IHC correlation from left supraclavicular lymph node / lung nodule a left lower lobe.
- Other findings as described above.

Dr. Gayana S M.D. (PGIMER), FEBNM
 Consultant, Nuclear Medicine and PET-CT

Dr. Raksha Kulkarni, DNB
 Senior Resident, Nuclear Medicine and PET-CT



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F18 FDG PET CT SCAN**TECHNIQUE:**

CT Protocol: Contrast enhanced CT scan was obtained from head to mid-thigh with Multislice CT scanner. Whole-body smart 3D modulation mode acquisition was adopted to achieve ALARA.

PET Protocol: With the patient fasting for 6 hours, 6 mCi of FDG was injected intravenously and 3D PET scan was performed 50 minutes later on Discovery IQ PETCT scanner; images were reviewed after normal and Q Clear iterative reconstruction techniques.

Patient Parameters: FBS: 94 mg/dl Ser.Creat: 0.71 mg/dl

Indication: Renal lesion, for evaluation.

FINDINGS:

- An FDG avid well defined heterogeneously enhancing lesion is seen in the anterosuperior aspect of upper pole cortex of the left kidney, measuring 21 x 17 x 37 mm with Max SUV 11.9; a heterogeneously enhancing FDG avid asymmetrical circumferential mural thickening is noted is also noted in the pelvicalyceal system of left kidney extending to left ureter with maximum thickness of 7 mm with Max SUV 9.3; significant FDG avid stranding is seen surrounding left ureter and pelvicalyceal system with resultant left hydronephrosis with dilatation of calyces in upper pole; single left renal artery is seen; linear filling defect is seen in the left renal vein - likely thrombus; the IVC appears unremarkable.
- Simple cortical cyst is seen in the right kidney; right kidney and ureter are otherwise unremarkable.
- The left adrenal gland appears bulky and nodular with low-grade FDG avidity, Max SUV 2.6. The right adrenal gland appears unremarkable.
- Multiple FDG avid left renal hilar, left para-aortic, pre-aortic, aortocaval, paracaval and bilateral common iliac lymph nodes are seen, few of these nodes show necrosis and surrounding stranding largest lymph node measures about 12 x 19 mm with Max SUV 9.8.
- The liver measures about 13.4 cm craniocaudally; an FDG avid hypodense lesion is seen in the segment VII of the liver, measuring 12 x 13 mm with Max SUV 2.1; a small 4 mm simple cyst is seen in the segment II of the liver; rest of the liver appears unremarkable.
- Multiple FDG avid nodules are seen scattered in the bilateral lung fields, largest in the lower lobe of left lung, measuring 30 x 16 mm with Max SUV 10.4.
- A few FDG avid anterior mediastinal, prevascular, pre - paratracheal, AP window and right hilar lymph nodes are seen, largest paratracheal lymph node measuring 28 x 22 mm with Max SUV 11.2.
- An FDG avid left supraclavicular lymph node is seen, measuring about 8 x 9 mm with Max SUV 3.4.
- No abnormal enhancing or FDG avid lesions in the brain or the skeleton.

Other findings:

Physiological FDG concentration is seen in the brain, salivary glands, oral cavity, heart, GIT, kidneys and urinary bladder.

